

Kragujevac, Serbia

WMO: 132780

Lat: 44.033N

Lon: 20.933E

Elev: 185

StdP: 99.12

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.3	-12.8	1.3	-10.7	-10.3	1.6	-8.1	8.6	7.0	7.2	4.3	0.9	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	35.0	21.5	33.2	21.4	31.3	21.0	23.2	31.4	22.4	30.5	21.6	29.5	1.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.7	15.7	27.1	19.8	14.9	26.2	19.0	14.1	25.0	69.8	31.4	66.7	30.4	63.8	29.6	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.2	4.3	-16.1	36.8	4.3	1.9	-19.2	38.1	-21.8	39.2	-24.2	40.3	-27.4	41.7		
(5)			-16.4	24.9	4.3	1.0	-19.5	25.6	-22.0	26.1	-24.5	26.7	-27.6	27.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	1.3	3.4	7.9	12.9	16.7	21.0	22.8	23.0	18.2	12.5	8.4	3.0	(6)
(7)		DBStd	8.59	5.39	5.49	4.52	3.53	3.49	3.33	2.98	2.94	4.04	4.24	4.73	4.73	(7)
(8)		HDD10.0	895	272	191	93	12	1	0	0	0	1	24	83	220	(8)
(9)		HDD18.3	2529	529	418	322	165	73	13	3	2	50	183	297	474	(9)
(10)		CDD10.0	1863	1	6	29	101	208	329	396	402	248	102	36	4	(10)
(11)		CDD18.3	456	0	0	0	3	23	92	140	147	48	3	1	0	(11)
(12)		CDH23.3	5059	0	0	2	72	265	981	1552	1629	504	53	1	0	(12)
(13)		CDH26.7	2083	0	0	0	13	51	368	694	771	182	5	0	0	(13)
(14)	Wind	WSAvg	1.6	1.7	1.8	2.1	1.7	1.5	1.5	1.4	1.4	1.4	1.6	1.8	(14)	
(15)	Precipitation	PrecAvg	652	42	39	47	56	67	74	66	54	60	56	47	50	(15)
(16)		PrecMax	1012	94	80	122	142	230	152	271	131	147	119	107	126	(16)
(17)		PrecMin	391	9	11	3	4	20	25	11	0	12	1	3	6	(17)
(18)		PrecStd	141	23	19	32	30	43	32	50	35	34	33	25	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	19.9	22.8	28.5	30.3	34.4	37.3	37.6	34.3	26.9	22.2	17.3	(19)
(20)			MCWB	10.4	12.0	13.6	17.5	19.5	22.5	21.6	22.1	21.0	17.5	15.2	12.3	(20)
(21)		2%	DB	12.7	16.4	20.6	25.7	28.0	32.4	34.9	35.2	31.4	24.9	20.0	14.6	(21)
(22)			MCWB	8.8	10.7	12.1	16.4	19.0	22.3	21.6	21.5	19.7	16.8	14.0	10.5	(22)
(23)		5%	DB	10.8	13.8	18.3	22.8	26.4	30.6	32.9	33.2	28.3	22.8	17.9	12.2	(23)
(24)			MCWB	7.7	9.2	11.3	14.5	18.4	21.5	21.5	21.1	18.9	15.9	12.9	8.9	(24)
(25)		10%	DB	8.7	11.5	15.7	20.6	24.5	28.7	30.9	31.0	26.2	20.3	15.9	9.8	(25)
(26)			MCWB	6.3	7.9	9.8	13.5	17.5	20.7	21.1	20.7	17.9	14.8	11.8	7.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.4	14.2	18.4	20.9	24.4	24.0	24.1	21.7	18.7	15.6	12.7	(27)
(28)			MCDWB	14.7	18.7	22.0	26.9	27.7	31.7	31.9	32.8	32.1	25.7	21.2	16.6	(28)
(29)		2%	WB	9.2	10.8	12.6	16.6	19.8	23.1	23.1	22.9	20.3	17.3	14.2	10.7	(29)
(30)			MCDWB	12.4	15.9	19.0	24.3	26.6	30.4	31.5	31.7	29.3	23.6	19.4	14.2	(30)
(31)		5%	WB	7.8	9.4	11.4	15.2	18.8	22.1	22.3	22.1	19.4	16.2	13.2	8.9	(31)
(32)			MCDWB	10.4	13.4	17.4	21.7	25.5	29.3	30.1	30.8	27.0	21.8	17.3	11.8	(32)
(33)		10%	WB	6.5	8.0	10.3	14.0	17.9	21.1	21.5	21.2	18.5	15.2	12.1	7.3	(33)
(34)			MCDWB	8.6	11.3	15.5	20.0	23.8	27.7	29.3	29.6	24.9	19.7	15.5	9.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	8.0	9.8	11.2	11.2	11.4	12.5	13.6	11.7	10.1	8.9	6.6	(35)
(36)			MCDBR	9.7	13.3	14.8	16.4	15.0	14.8	16.9	18.0	15.9	14.7	12.0	10.4	(36)
(37)			MCWBR	6.6	8.1	7.9	8.4	7.2	6.2	6.2	6.4	6.4	7.7	6.8	6.9	(37)
(38)		5% WB	MCDWR	9.4	12.2	13.3	14.5	13.6	13.5	13.7	15.0	13.9	13.4	11.0	9.6	(38)
(39)			MCWBR	6.5	7.6	7.4	7.6	6.9	6.2	5.9	6.0	6.1	7.2	6.4	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.340	0.366	0.402	0.460	0.450	0.443	0.444	0.457	0.434	0.394	0.356	0.336	(40)	
(41)		taud	2.332	2.235	2.201	2.059	2.149	2.211	2.208	2.170	2.204	2.308	2.364	2.357	(41)	
(42)		Ebn at Noon	766	806	826	804	826	832	825	799	784	767	741	731	(42)	
(43)		Edh at Noon	82	107	126	158	148	140	139	140	125	99	80	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.32	2.15	3.32	4.44	5.39	6.07	6.21	5.57	3.90	2.68	1.67	1.14	(44)	
(45)		RadStd	0.19	0.27	0.41	0.50	0.55	0.60	0.48	0.53	0.40	0.34	0.17	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		+0.67	N/A	N/A	N/A	+0.39	N/A	N/A	-156	+105	N/A

Nomenclature: See separate page